Adapter

Season: 2 · Structural · Category: GoF · Level: Beginner

Intent

- Convert one interface into another clients expect.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- A travel plug adapter between US and EU sockets.

When to Use

- See Intent.
- The design force named in Intent shows up more than once.

When Not to Use

- When a simpler redesign removes the need.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.gof.structural.adapter;

interface JsonLogger {
    void logJson(String json);
}

final class LegacyXmlLogger {
    String last;
    void logXml(String xml) { last = xml; }
}

/** Object adapter — wraps adaptee, translates calls. */
public final class XmlToJsonLoggerAdapter implements JsonLogger {
    private final LegacyXmlLogger legacy = new LegacyXmlLogger();

    @Override
    public void logJson(String json) {
        legacy.logXml("<payload>" + json + "</payload>");
    }

    String lastXml() { return legacy.last; }
}
```

Modern Java 25 / Functional

- Prefer sealed interfaces + records; virtual proxy uses synchronized lazy init.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: gof/structural/adapter/
```

Pitfalls

- Keep adapters thin.
- Decorators should preserve the interface.
- Facades must not become god objects.

Interview Q&A

- Q: When is Adapter the wrong choice? A: When you can change the adaptee/subject API or collapse the hierarchy.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Convert one interface into another clients expect.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: A travel plug adapter between US and EU sockets. [0:40] Intent — State the design force in one sente...